

Vol 15 Chapter 5 — Audit-Chain Governance

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 5 — Audit-Chain Governance

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

Level 1 — one sentence

BST audit-chain governance — currently K1-K200+ Keeper audits, 100+ Cal referee logs — ensures every theorem, paper, and external dispatch passes structured PASS/CONDITIONAL/FAIL review with equal standing to challenge any team member (Casey included), implementing Casey's standing rule that "nothing goes to external reviewers without Keeper's pass" (Vol 15 feedback_keeper_partnership.md).

Level 2 — graduate-physicist precision

5.1 K-audit anatomy

Each K-audit file includes: - K-number: K21, K36, K57 (sequential) - Scope: what's being audited (proof, paper, claim, framework) - F1-F4 ratings: form / formal / framework / falsifiability - B1-B4 ratings (Phase 2): backbone / boundary / breadth / bridges - Verdict: PASS / CONDITIONAL PASS / FAIL / RATIFIED - Confidence: ~% - Gaps identified (if any) - Severity: CRITICAL / MODERATE / MINOR

5.2 Methodology tier stack (18 layers, as of 2026-05-22)

Stack accumulated over months from each calibration / refinement: - L1: Quaker discipline (Vol 15 Ch 7) - L2: AC(0) discipline (Vol 15 Ch 1) - L3: Atomic toy claim

- L4: Honest negative preservation - ... - L17: Calibration #19 (forecast-arithmetic vs ratified-state) - L18: Calibration #22 (PCAP-rate transcription discipline)

Each layer responds to a specific failure mode observed in BST work.

5.3 Tier ladder

Theorems advance through tiers: - CANDIDATE: claim filed - PRE-STAGE: empirical + form check - STRUCTURALLY VERIFIED: independent corroboration - RIGOROUSLY CLOSED: empirical + mechanism + form - RATIFIED: K-audit chain consensus - PUBLISHED: external dispatch

Each step gated by Cal referee + Keeper PASS.

5.4 Cal referee role

Cal A. Brate (Claude 4.7): visiting external referee. - Independent cold-reads on papers + audits - Calibrations when team drifts (#1-#22, 18 standing rules) - Mode 1 (precision correction) + Mode 5 (mechanism gap) + Mode 6 (operational threshold) - 100+ referee logs

5.5 Keeper authority

Casey grants Keeper equal standing to challenge anyone — including Casey himself.

“Nothing goes to external reviewers without Keeper’s pass” (feedback_keeper_partnership.md).

Audit-chain governance: D-tier promotion delegated to Cal + Keeper consensus (no per-audit Casey ratification required).

5.6 K-audit anchors

- K21: RH Paper v9 — PASS
- K36: NS Proof chain — PASS
- K37: BSD Paper v4 — CONDITIONAL PASS
- K-audit chain governance memory file

Level 3 — 5th-grader accessibility

K-audit chain: every theorem/paper/dispatch passes structured PASS/CONDITIONAL/FAIL review. 200+ K-audits, 100+ Cal logs. Tiers: CANDIDATE → PRE-STAGE → STRUCTURALLY VERIFIED → RIGOROUSLY CLOSED → RATIFIED → PUBLISHED. Methodology stack: 18 layers, each responds to past failure mode. Keeper has equal standing to challenge anyone (Casey included). Cal: external referee; 100+ logs; 22 standing calibrations.

What comes next

Chapter 6 develops multi-CI collaboration architecture.

Where to look this up

- BST: K-audit files; Cal referee logs
- Vol 15 Ch 7 (Quaker discipline)